Introduction

Deque
Data Structures and its Applications (UE25CS52A)

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Definition and Properties

- A **deque**
  - Is a data structure
  - Supports add and delete
    - At both the ends
- Pronounced as deck

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Implementation of stack using deque

- Achieved by composition
- Also called embedding|layering|containment
- Is a reuse mechanism
- Avoids reinventing the wheel

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Implementation of stack using deque

- **Deque interface**:
  - void init(dll_t* ptr_dlist);
  - void disp_forward(dll_t* ptr_dlist);
  - void disp_backward(dll_t* ptr_dlist);
  - void add_in_begin(dll_t* ptr_dlist, int key);
  - void deinit(dll_t* ptr_dlist);
  - void add_at_end(dll_t* ptr_dlist, int key);
  - int remove_in_begin(dll_t* ptr_dlist);
  - int remove_at_end(dll_t* ptr_dlist);
  - int is_empty(dll_t *ptr_dlist);
  - int is_full(dll_t *ptr_dlist);

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Implementation of stack using deque

- **Stack interface**:
  - void push(stack_t* ptr_stack, int key);
  - int pop(stack_t* ptr_stack);
  - int is_empty_stack(stack_t* ptr_stack);
  - int is_full_stack(stack_t* ptr_stack);
  - void init_stack(stack_t* ptr_stack);
  - void deinit_stack(stack_t* ptr_stack);

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Implementation of stack using deque

- **Deque interface mapping for stack interface**
  - void push(stack_t* ptr_stack, int key);
    => void add_in_begin(dll_t* ptr_dlist, int key);
  - int pop(stack_t* ptr_stack);
    => void add_in_begin(dll_t* ptr_dlist, int key);
  - int is_empty_stack(stack_t* ptr_stack);
    => int is_empty(dll_t *ptr_dlist);
  - int is_full_stack(stack_t* ptr_stack);
    => int is_full(dll_t *ptr_dlist);
  - void init_stack(stack_t* ptr_stack);
    => void init(dll_t* ptr_dlist);
  - void deinit_stack(stack_t* ptr_stack);
    => void deinit(dll_t* ptr_dlist);

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deque

Stack functions delegate to the deque functions

```
void push(stack_t* ptr_stack, int key)
{
        add_in_begin(&ptr_stack->list_, key);
}
int pop(stack_t* ptr_stack)
{
        return remove_in_begin(&ptr_stack->list_);
}
```


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THANK YOU

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